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Studierichting: Informatica

Oefeningenreeks 6A nr. 9.6

a, b, c, d

Rekenkundige rij met $ad = 54, bc = 104$

$$\Rightarrow b = a + v, c = a + 2v, d = a + 3v$$

$$\Rightarrow a(a + 3v) = 54, (a + v)(a + 2v) = 104$$

$$\Rightarrow a^2 + 3av = 54, a^2 + 3av + 2v^2 = 104 \Rightarrow 54 + 2v^2 = 104 \Rightarrow v = \pm 5$$

$$\Rightarrow a^2 \pm 15a - 54 = 0 \Rightarrow D = 15^2 + 4 \cdot 54 = 21^2$$

$$v = 5 \Rightarrow a \in \{3, -18\}, v = -5 \Rightarrow a \in \{-3, 18\}$$

$$\Rightarrow 3, 8, 13, 18 \text{ en } 18, 13, 8, 3 \text{ en } -3, -8, -13, -18 \text{ en } -18, -13, -8, -3$$

References